

Designing a database for a Virtual Learning Environment

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Introduction

This report details a proposal for a database to support a Virtual Learning Environment where university staff and students can engage in courses – holding teaching sessions, posting reading materials and assignments, uploading authored work, posting grades achieved, and more.

The report will discuss the design of an Entity-Relation diagram with reasoning behind it, after which a subsequent logical model and the methods it was derived with will be presented.

The conceptual model

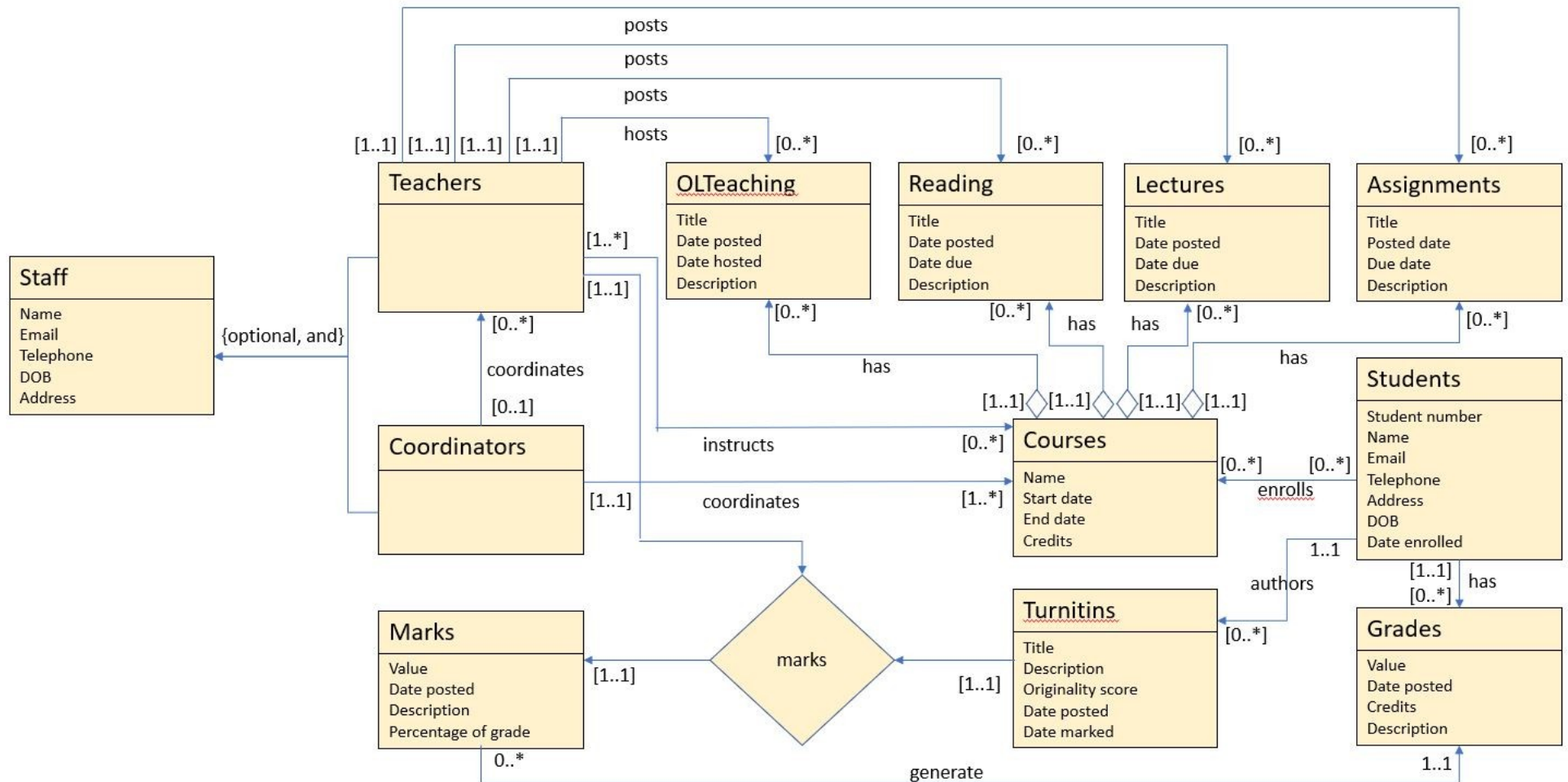


Figure 1: the ER diagram for the proposed VLE database

Account of assumptions and decisions made when constructing the conceptual model

To lay a foundation for the model, all relevant entities were identified: the people involved, which implied staff in two capacities and students, courses and their component materials, Turnitin materials (assignment products authored by the student), and the resulting marks and grades. These were then placed in a table where relations between them were examined, in both directions. The result of this can be seen in Table 1 on the next page. Although more relations could be suggested, after review most of those were derived from the relations that are now shown in the table: for instance, an assignment is assigned to the student, but as the assignment is a part of the course and the student is enrolled in this course, it is implied that the student is assigned this task.

The table shows all binary relations. There is one ternary relationship in the diagram, between Turnitins, teachers, and marks. A teacher cannot generate a mark, nor can a Turnitin file, it requires both to generate data for the mark table. This has been shown by a diamond linking all three entities.

After the relationships had been modelled, multiplicity has been added, a single ISA relationship was drawn, and aggregation relationships were defined. This has been done on a few assumptions:

- teachers need not be instructing a course (they might be engaged with other work such as developing skills)
- every course must have one coordinator and at least one teacher
- a coordinator will oversee at least one course and any number of teachers
- a member of staff can be a teacher, coordinator, neither or both; and in the last case not necessarily for the same course (in the case they are neither they might for instance be temporarily unassigned)
- a member of staff need not be either a teacher or coordinator
- course materials are always posted by one teacher
- online teaching sessions are hosted by a single teacher
- a course can contain any number of reading materials, lectures, assignments, and teaching sessions, and those are always contained within a course (aggregation)
- a course can have any number of students enrolled and a student can be enrolled in any number of courses (a course with no students might have just been opened and is still awaiting enrolments)
- all Turnitin materials have a single author, are marked by a single teacher, and have a single mark as a result
- marks have a weighting percentage and together generate a grade. A grade can consist of one or more marks and is only generated at the end of the course, so has a different posting date.

The ISA relationship was defined as {optional / and}, for reasons given above.

Lastly all the attributes of the entities were defined and are shown in table 2 below. There are no attributes for teachers and coordinators as yet; but because of their overlapping roles yet functional interrelationship and the possibility of adding specific tasks and relations to them in the future they were modelled as subclasses and separate entities rather than referred to by staff numbers.

	Students	Staff	Teachers	Coord.	Courses	Assign.	Lectures	Reading	OLTeaching	Turnitins	Marks	Grades
Students					Enrols					Authors		Has
Staff												
Teachers		Subcl. of			Instructs	Posts	Posts	Posts	Hosts		Marks	
Coordinators		Subcl. of	Coord.s		Coordinates							
Courses						Has	Has	Has	Has			
Assignments												
Lectures												
Reading												
OLTeaching												
Turnitins											Marks	
Marks												Generates
Grades												

Table 1: the binary relationships between entities. The table is read starting at row names, relationships with columns headings are defined at their intersection.

Students	Staff	Teachers	Coord.	Courses	Assignments	Lectures	Reading	OLTeaching	Turnitins	Marks	Grades
Student ID	Name			Name	Title	Title	Title	Date posted	Title	Value	Value
Name	Email			Start date	Posted date	Date posted	Date posted	Date hosted	Originality score	Date posted	Date posted
Email	Tel			End date	Due date	Date due	Date due	Description	Date posted	Grade perc.	Credits
Telephone	Address			Credits	Mark %	Description	Description		Description	Description	Description
Address	DOB				Description						
DOB											
Date enr.											

Table 2: the attributes of the entities

The logical model

The logical model for this database is shown below.

- Students (studentNo, studentID, Fname, Lname, email, telephone, address1, address2, city, postcode, DOB, dateEnrolled)
Primary Key studentNo
- Staff (staffNo, Fname, Lname, email, telephone, address1, address2, city, postcode, DOB)
Primary Key staffNo
- Teachers (teacherNo, coordinatorNo)
Primary Key teacherNo
Foreign Key coordinatorNo references Coordinators(coordinatorNo)
- Coordinators (coordinatorNo, coursesCoordinating, teachersCoordinating)
Primary Key coordinatorNo
- Courses (courseNo, name, startDate, endDate, credits, coordinatorNo)
Primary Key courseNo
Foreign Key coordinatorNo references Coordinators(coordinatorNo)
- Assignments (assignmentNo, name, postedDate, dueDate, markPercentage, description, teacherNo, CourseNo)
Primary Key assignmentNo
Foreign Key teacherNo references Teachers(teacherNo)
Foreign Key courseNo references Courses(courseNo)
- Lectures (lectureNo, title, datePosted, dateDue, description, teacherNo, courseNo)
Primary Key lectureNo
Foreign Key teacherNo references Teachers(teacherNo)
Foreign Key courseNo references Courses(CourseNo)
- Reading (readingNo, title, datePosted, dateDue, description, teacherNo, courseNo)
Primary Key readingNo
Foreign Key teacherNo references Teachers(teacherNo)
Foreign Key courseNo references Courses(CourseNo)
- OLTeaching (olteachingNo, datePosted, dateHosted, description, teacherNo, courseNo)
Primary Key olteachingNo
Foreign Key teacherNo references Teachers(teacherNo)
Foreign Key courseNo references Courses(courseNo)
- Turnitins (turnitinNo, title, author, description, originalityScore, datePosted, studentNo)
Primary Key turnitinNo
Foreign Key studentNo references Students(studentNo)

- Marks (markNo, value, datePosted, gradePercentage, description, gradeNo)
Primary Key markNo
Foreign Key gradeNo references Grades(gradeNo)
- Grades (gradeNo, value, datePosted, credits, description, studentNo)
Primary Key gradeNo
Foreign Key studentNo references Students(studentNo)
- Enrolment (studentNo, courseNo)
Primary Key studentNo, courseNo
Foreign Key studentNo references Students(studentNo)
Foreign Key courseNo references Courses(courseNo)
- Instructors (teachNo, courseNo)
Primary Key teachNo, courseNo
Foreign Key teachNo references Teachers(teachNo)
Foreign Key courseNo references Courses(courseNo)
- Marking (turnitinNo, teachNo, gradeNo)
Primary Key turnitinNo, teachNo, markNo
Foreign Key turnitinNo references Turnitins(turnitinNo)
Foreign Key teachNo references Teachers(teachNo)
Foreign Key gradeNo references Marks(markNo)

Account of assumptions and decisions made when constructing the logical model

To generate the logical model, first all the entities and their attributes were written as tables and given an unique primary key which was always the name of the table follow by No. StudentID, although unique, was not used as it is still dependent on the universities administration and IDs might not be issues in time, change or have unusual formats. In this database, a student that enrolls is issued an identifier dedicated to the database.

After the tables were written all the multiplicities of the conceptual model were gathered into a table of relations, the result of which can be seen below:

	Stud.	Staff	Teach.	Coords	Courses	Assigns	Lects	Read	OL teach	TitIns	Marks	Grades
Stud												
Staff												
Teachs												
Coords			1:*									
Cours	*.*		*.*	1:*								
Assigns			1:*		1:*							
Lects			1:*		1:*							
Read			1:*		1:*							
OLTs			1:*		1:*							
Turns	1:*		1:1									
Marks	1:*		1:1							1:1		
Grades											1:*	

Table 3: binary relationships in the database. Some entity names have been abbreviated to fit the table on the page.

The 1:* relationships were added to the tables by including the primary key of the parent in the table of the child as a foreign key.

The *.* relationships were modelled in a table of their own, Enrolment and Instructors, where the primary key consisted of both the primary keys of the two entities in the relationship, and both those keys were also added as foreign keys. Any unique combination of the two is now a tuple which refers to both entities.

The 1:1 are part of a ternary relationship between Turnitins, Teachers and Grades

A list of recommendations supporting the databases implementation

To implement the database, data access restrictions must be set in place for various users.

Students, teachers and coordinator must have different access levels, to prevent students from accessing materials such as test answers, teachers accessing materials such as students' notes, or coordinators bundling data and disseminated it to audience with limited access of their own.

Manipulation access must also be limited to for instance prevent students from changing their uploaded assessed materials or preventing teachers from changing course materials in active use.

Reference list of the key sources and resources used

- Course materials provided for Database Systems and Big Data, CS5097, CS5098, University of Aberdeen

Appendix 1: Data dictionary

Below is the data dictionary with definitions of all entities and attributes. Relationships are defined by their multiplicity and binary, ternary or higher nature, and their description, all of which has been included in the body of the report.

The field size has been kept short where possible such as for numbers of known length, and given enough space for unknown data such as text for international contact details to be entered without error.

The description attributes have been filled with 'Lorem Ipsum', which signifies that any text can be entered into these fields that the content creator requires.

Table	Field name	Data type	Field size	Description	Example
Students	studentNo	Integer	8	Database ID of student	0001
	studentID	Integer	8	University ID of student	12345678
	Fname	Text	30	First name of student	Robert
	Lname	Text	30	Last name of student	Burns
	email	Text	50	Email of student	Robert.Burns@lit.ac.uk
	telephone	Text	20	Telephone number of student	+447712345678
	address1	Text	50	First line of address, house name or street number	The Tower
	address2	Text	50	Second line of address, additional information	Cell Umpteen
	city	Text	30	City of address	London
	country	Text	30	Country of address	United Kingdom
	postcode	Text	10	Postcode of address	EC3N 4DR
	DOB	Date/time	10	DOB of student	25/01/1759
	dateEnrolled	Date/time	10	Date when student enrolled at university	15/01/2020
	staffNo	Integer	8	Database ID of staff member	0002
Staff	Fname	Text	30	First name of staff member	Robert
	Lname	Text	30	Last name of staff member	Burns
	email	Text	50	Email of staff member	Robert.Burns@lit.ac.uk
	telephone	Integer	11	Telephone number of staff member	07712345678
	address1	Text	50	First line of address, house name or street number	The Tower
	address2	Text	50	Second line of address, additional information	Cell Umpteen
	city	Text	30	City of address	London
	postcode	Text	6	Country of address	United Kingdom
	country	Text	30	Postcode of address	EC3N 4DR
	DOB	Date/time	10	DOB of staff member	25/01/1759
	teachNo	Integer	8	Database ID of teacher	0002
	coordNo	Integer	8	Database ID of coordinator	0003
Teachers Coordinators					

Courses	courseNo	Integer	8	Database ID of course	0004
	name	Text	50	Name of the course	Database Systems
	startDate	Date/ time	10	Date when course started	15/01/2020
	endDate	Date/ time	10	Date when course ends	15/03/2020
Assignments	credits	Integer	2	University credits associated with course	15
	assignmentNo	Integer	8	Database ID of assignment	0005
	name	Text	50	Title of the assignment	Practical 1, XAMPP tutorial
	postedDate	Date/ time	10	Date when assignment was posted	15/01/2020
Lectures	dueDate	Date/ time	10	Date when assignment is due for completion	15/03/2020
	markPercentage	Integer	2	Weighting towards final grade for the course	15
	description	Text	204	The contents of the assignment or links to the content	15/03/2020
	lectureNo	Integer	8	Database ID of lecture	Lorem ipsum
Reading	title	Text	50	Title of the lecture	0006
	datePosted	Date/ time	10	Date when lecture was posted	Lecture 1, Course Overview
	dateDue	Date/ time	10	Date when lecture is due for review by the student	15/01/2020
	description	Text	204	The contents of the lecture or links to the content	15/03/2020
Online Teaching	readingNo	Integer	8	Database ID of reading material	0007
	title	Text	50	Title of the reading materials	Chapter 1
	datePosted	Date/ time	10	Date when reading materials were posted	15/01/2020
	dateDue	Date/ time	10	Date when reading materials are due for review by the student	15/03/2020
Turnitins	description	Text	204	The reading materials or links to them	15/03/2020
	olteachingNo	Integer	8	Database ID of online teaching session	Lorem ipsum
	datePosted	Date/ time	10	Date when online teaching session was announced	0008
	dateHosted	Date/ time	10	Date when teaching session is hosted	15/01/2020
Turnitins	description	Text	204	Details about the teaching session including time and duration	15/03/2020
	turnitinNo	Integer	8	Database ID of Turnitin	0009
	title	Text	50	Title of the Turnitin materials	Assessment 1 report
	authorFname	Text	30	First name of the author of the Turnitin materials	Robert
	authorLname	Text	30	Last name of the author of the	Burns

Marks			204	Turnitin materials	
	description	Text	8	Description of the Turnitin materials by the author	Lorem ipsum
	originalityScore	Integer	3	Result of the originality tester web application embedded in the VLE	100
	datePosted	Date/time	10	Date the Turnitin materials were posted	15/01/2020
	dateMarked	Date/time	10	Date the Turnitin materials were graded	15/03/2020
	markNo	Text	8	Database ID of mark	0010
	datePosted	Date/time	10	Date the mark was posted	15/01/2020
Grades	gradePercentage	Integer	3	Percentage of course grade that the mark contributes	25
	description	Text	204	Description of how the grade was derived from the Turnitin materials	Lorem ipsum
	gradeNo	Text	8	Database ID of grade	0010
	value	Text	2	The value of the grade	A1
	datePosted	Date/time	10	Date the grade was posted	15/01/2020
	description	Text	204	Description of how the grade was derived from the Turnitin materials	Lorem ipsum